

FHIR + AI Hackathon - Session 1 Orientation

Session 1: FHIR Fundamentals with LLM-Assisted Code Generation

Duration: 1 hour

Today's Clinical Scenario

Find patients with Type 2 diabetes, retrieve their most recent HbA1c values, and identify those with poor glycemic control (HbA1c > 7.5%).

This is a real clinical informatics task that requires:

What is FHIR?

Fast Healthcare Interoperability Resources (FHIR)

- Standard for exchanging healthcare information electronically

Server we're using:

Public sandbox with synthetic patient data (Synthea)

FHIR Resources You'll Use Today

Condition

Patient

Observation

Clinical Codes Reference

| Code | System | Meaning | Reference |

HbA1c Interpretation:

How FHIR Resources Link Together

Condition (SNOMED CT: 44054006 - Type 2 Diabetes)

To answer our question, we must:

FHIR Query Examples

Search for Conditions

Get a Patient

Search for Observations

Your Workflow Today

Step 1: Search Conditions

Step 2: Get Patient Demographics

Step 3: Get Lab Values

Step 4: Analyze Results

Step 5: Generate Summary

Key Learning Points

1. LLM as Code Generator
2. Multi-Step Clinical Queries
3. Separation of Concerns

What to Expect

[OK] Success looks like:

[!] Common issues:

Tips for Working with Claude

Good Prompt

Poor Prompt

Be specific: Include the code system, exact codes, and desired output format

Next Session Preview

Session 2: Tool Use & Agents

Today you:

Session 2:

This is called "tool use" or "function calling"

Questions?

Before you start:

Let's begin!

Resources

- FHIR Server: <https://launch.smarthealthit.org/v/r4/fhir>

Remember: Use SNOMED CT codes (not ICD-10) for condition searches!